

MUHAMMAD FAIZAN AHMED

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Summary

MSc Robotics & Automation student at Manchester Metropolitan University with hands-on experience in PLC programming, SCADA development, and industrial automation system integration. Experienced in designing and implementing control and monitoring solutions for production environments. Strong Interest in robotics and automation engineering roles involving system design, integration, testing, and industrial deployment.

Portfolio: <https://engr-muhammad-faizan-ahmed.github.io/portfolio/>

Skills

- **Industrial Automation & Control:** PLC Programming (Siemens, Delta), SCADA (WinCC Unified), HMI Development, Virtual Commissioning
 - **Robotics & Autonomous Systems:** Industrial Robotics, Cobots Coordination, Robot Simulation (RoboDK, CoppeliaSim), ROS, Robot Kinematics & System Integration
 - **Computer Vision & AI:** Machine Vision (OpenCV, YOLO), Industrial Inspection Systems
 - **Programming:** Python, C++, MATLAB, Lua, Embedded Programming
 - **IIoT & Smart Manufacturing:** Sensors Integration, Node-RED, Siemens InsightsHub, Senseye, Industrial Data Monitoring
 - **Electronics & Hardware Development:** PCB Design (Altium, EasyEDA), Embedded Systems, Prototyping
 - **Mechanical Design & Digital Manufacturing:** 3D Modelling (NX CAD, SolidWorks, Fusion 360), Additive Manufacturing (3D Printing)
 - **Professional Skills:** Problem-Solving, Analytical Thinking, Technical Documentation, Multidisciplinary Teamwork, Project Ownership
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Education

MSc Robotics & Automation

Manchester Metropolitan University | Sep 2025 – Sep 2026 (Expected)

MSc Dissertation - Cobot Synchronisation: Coordinated control of collaborative robots for industrial sewing task optimisation

Projects:

- UR20 Robotic Sorting & Packaging Workcell
- PLC Automation & Virtual Commissioning System (Selected for Siemens Transform 2026)
- Industry 4.0 Environmental Monitoring System
- Phones-R-Us Industry 4.0 Smart Factory
- Warehouse Robot Navigation Using AI Search Algorithms

Bachelor of Engineering in Mechatronics Engineering

Mehran University of Engineering & Technology | Oct 2019 – Dec 2023

BE Dissertation - Design and Development of Soft Robotics Gripper for Dexterous Grasping

Professional Experience

Automation Engineer

Industronics Engineering | Mar 2025 – Aug 2025

- Designed and implemented a full WinCC Unified SCADA system for a fully automated warehouse for a UK-based client.
- Designed and deployed a vision-based cloth width measurement system using computer vision for real-time industrial textile inspection.
- Applied industrial control, machine vision, and system integration skills in production-focused automation projects.

Hardware Design Engineer

INTEL Lab, PNEC NUST | Mar 2024 – Mar 2025

- Worked on the design, prototyping, and testing of drones, including quadcopters and high-speed targeted drones.
- Contributed to a radar simulator project by developing a user-friendly frontend interface using PyQt6 and integrating software components in Linux environments.
- Developed a product utilizing LoRaWAN communication for long-range, low power IoT applications, ensuring reliable data transmission in diverse environments.

IoT Engineer

Indus Technetronic | Jul 2023 – Dec 2023

- Developed a home automation smart switching IoT device ("Beetle") for a startup, supporting hardware design and system integration.
- Worked on a residential load consumption monitoring system, enabling real-time energy usage analysis.
- Designed PCBs and 3D-printed enclosures and supported end-to-end device prototyping.

Certification

- Professional Industrial Automation
- Universal Robots e-Series Core Track
- Universal Robots e-Series Pro Track
- Universal Robots e-Series Application Track
- Programming for Everybody (Getting Started with Python)
- Introduction to Mechanical Engineering Design and Manufacturing with Fusion 360
- Introduction to IIoT and Industry 4.0
- AI for Everyone
- Neural Networks and Deep Learning

Achievement

- Gold & Silver Medalist
- Selected for Siemens Transform 2026 showcase
- Awarded Allan & Nesta Ferguson Masters Scholarship for academic excellence